FORCE

TIME 1HR **MARKS:40**

SECTION A

[*Attempt **ALL** questions in this section*]

Question 1: **[5]**

[i] Moment of force acting at the pivot vertically downwards is …….

[a] maximum

[b] minimum

[c] zero

[d] infinite.

[ii] Two or more forces acting on a circular lamina not pivoted produces…..

[a] translational motion

[b] rotational motion

[c] both translational and rotational motion

[d] none of the above

[iii] When a car moves along a ghat section the outward seeking force is….

[a] due to inertia of motion

[b] due to centrifugal force

[c] due to inertia of direction

[d] both b and c

[iv] Centre of gravity of a hit me doll is ……

[a] towards its base

[b] at the geometric centre

[c] towards its top

[d] none of the above

[v] Principle of moments is applicable for……

[a] flexible beam

[b] non-pivoted body

[c] couple

[d] uniform beam

Question 2: **[5]**

[i] Force towards the right of fulcrum acting upwards are considered to produce…… [positive / negative] torque.

[ii] Centre of gravity for a boomerang lies ……. [inside / outside] the object.

[iii] On reducing the speed of motorcycle the motorcyclist tends to come towards the centre of the circular path due to ……. [Centrifugal / Centripetal] force.

[iv] Longer the size of the key ……. [easier / difficult] it is to open a safe vault.

[v] Force necessary to produce uniform circular motion is ……. [Centripetal / Centrifugal] force.

Question 3:

[a] State two conditions for a body to be in equilibrium. **[2]**

[b] Where does the centre of gravity for a hollow cone lie from its apex on its vertical axis? **[2]**

[c] Two forces 20gf and 12gf acting at distance of 6cm and 10cm respectively from the turning point of a half meter scale. Show by suitable calculations whether the rod is in equilibrium or not. **[3]**

[d] A meter scale is balanced on a knife edge at its centre. 10g weight is placed one on top of the other at 12cm mark, the scale is found to be balanced at 45cm mark. Calculate the mass of the meter scale. **[3]**

SECTION B

[Attempt ***any two*** questions out of three**]**

Question 4:

[a] A uniform meter scale of mass 60g carries masses of 20g,30g and 80g from points 10cm,20cm and 90cm marks respectively. Where must the scale be pivoted for it to be in equilibrium? **[4]**

[b] Differentiate between: [i] Moment of force and Moment of Couple

[ii] Centripetal force and Centrifugal force **[4]**

[c] State the position of centre of gravity for: **[2]**

[i] Hollow sphere with thickness

[ii] Solid cylinder

Question 5:

[a] State the sign conventions used for applying principle of moments when different forces act towards the left or right of the fulcrum for the body to be in a state of equilibrium. **[4]**

[b] Where does the CG of following objects lie: **[3]**

[i] Isosceles trapezium

[ii] Cuboid

[iii] Annular ring.

[c] The following figure shows a window aircon that is attached to the wall. Due to its weight it tends to sag outwards. To prevent it from sagging a truss member is supporting the base of the aircon with a vertical force of 50N. Find the weight of aircon? Given width of the cage outside the wall is 1.5m **[3]**



Question 6:

[a] Two parallel unlike forces of 10N and 25N act at 12cm from each other. Find the point about which the body would balance. **[4]**

[b] A uniform meter scale is balanced at 40cm mark when weights of 20gf and 5gf are suspended at 5cm mark and 75cm mark respectively. Calculate the weight of meter scale. **[3]**

[c] A wheel of diameter 1.6m with axle at its geometric centre is applied with a force of 4N at point B tangential to the circumference of the wheel. Determine the moment of force about a point A situated at its diametrically opposite end on the periphery of the wheel. If the same force acts vertically along the axle would the moment of force be maximum or minimum? **[3]**